

Sri Lanka Institute of Information Technology



Machine Learning Based Campaign Response Prediction for Bank Marketing

Initial Group Submission

Machine Learning | IT3091

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Declaration

We hereby declare that this Initial Group Submission for the IT3091 Machine Learning module has been prepared by the undersigned group members. The business problem, primary decision lens, proposed machine learning task, workflow, and allocation of responsibilities presented in this report represent the proposed direction of our group project.

The project will use the Bank Marketing dataset supplied through the UCI Machine Learning Repository under the Guided Data Track. Information obtained from the assignment descriptor, dataset documentation, academic literature, software documentation, and any other supporting source will be acknowledged appropriately.

The group agrees to maintain accurate records of important decisions, data-quality findings, exploratory insights, preprocessing choices, experiments, evaluation results, individual contributions, and the use of Artificial Intelligence tools. Any AI-assisted material will be reviewed, verified, and declared in accordance with module requirements.

We understand the importance of academic integrity, reproducibility, responsible data use, and honest reporting. We will not present model performance without the evidence needed to explain how the data were split, how leakage was controlled, how models were compared, and how conclusions were reached.

Abstract

Direct marketing allows banks to offer products to selected customers, but telemarketing campaigns can consume considerable staff time and create unnecessary customer contact when response rates are low. The practical decision is therefore not simply whether a campaign will succeed overall, but which customer contacts should receive priority before the call is made.

This project proposes a machine-learning approach for campaign response prediction using the UCI Bank Marketing dataset. The primary task will be formulated as supervised binary classification, where the target indicates whether a contacted customer subscribed to a term deposit. The model will produce a probability of positive response for each campaign record. These probabilities can support customer ranking and contact-capacity decisions while preserving Campaign Response Prediction as the project's single primary lens.

The project will compare a no-skill baseline with Logistic Regression and at least three nonlinear alternatives. Evaluation will emphasise precision-recall area under the curve, recall, precision, F1-score, balanced accuracy, probability quality, and business-oriented measures such as lift and the proportion of responders captured within the highest-ranked customers. The final test portion will remain separate from model selection and threshold tuning.

Particular attention will be given to class imbalance, categorical values recorded as unknown, repeated campaign contacts, temporal generalisation, and target leakage. Call duration will be excluded from the realistic pre-contact model because it is only known after a call has occurred. The final recommendation will balance predictive performance, interpretability, campaign value, fairness, and limitations, including the dataset's historical Portuguese context and the difference between predicting response and proving that a marketing call caused it.

Acknowledgement

We would like to express our sincere appreciation to the lecturers and academic staff of the IT3091 Machine Learning module for providing the assignment framework, business scenario, and guidance required to begin this project. We also acknowledge the UCI Machine Learning Repository and the researchers associated with the Bank Marketing dataset for making the data and supporting documentation available for educational and research use. Finally, we thank all four group members for their planned collaboration, peer review, and shared responsibility throughout the project.

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1. Introduction

1.1 Background

Banks use direct marketing to introduce financial products, retain customer relationships, and generate deposits. Telephone campaigns can provide a direct channel for explaining an offer, but each call uses staff time and may create a poor customer experience when the customer has little interest in the product. A campaign manager must therefore decide how limited contact capacity should be distributed across eligible customers.

The assignment scenario states that a bank wants to improve campaign effectiveness and use customer information more intelligently [1]. The assignment scenario states that a bank wants to improve campaign effectiveness and use customer information more intelligently [1]. The selected UCI dataset records direct marketing contacts made by a Portuguese banking institution.

1.2 Problem Statement

A campaign that has a persistent low conversion rate and high operational cost associated with untargeted direct marketing. Calling clients at random leads to severe resource misallocation, inflates tele-agent labour expenditures, and risks damaging the bank's reputation through customer contact fatigue. The marketing department currently lacks a systematic, data-driven mechanism to qualify and prioritize leads before deploying human agents.

The central problem is therefore: How can customer, campaign, previous-contact, and contextual information available before a call be used to estimate the probability that the customer will subscribe to a term deposit?

However, an apparently accurate model can still be unsuitable if it relies on information unavailable before the decision, hides poor performance on the minority response class, or is evaluated using a split that leaks time-related patterns

1.3 Proposed Solution

We propose implementing a machine-learning decision-support workflow that assigns each eligible campaign record a probability of positive response. The bank could use the probabilities to rank candidates, select a contact threshold or top proportion consistent with available staff capacity, and compare the expected concentration of responders with an unprioritized campaign. The project will not automate customer contact or make decisions about credit eligibility.

2. Track Declaration

2.1 Selected track

The considered assignment track is a ‘guided data track’ with the assigned domain being ‘banking and finance’ and the assigned dataset being [Bank Marketing](#), which documents direct outbound telemarketing initiatives conducted by a retail banking institution offering long-term deposits.

The scope of this controlled investigation is confined to operational campaign efficiency, data-driven customer targeting, and predictive response modelling. Rather than treating marketing outreach as an unselective bulk tele-calling exercise, the project team will establish an evidence-based pipeline that systematically identifies, qualifies, and prioritises viable prospects prior to contact initiation.

2.2 reason for the chosen track

The Guided Data Track was chosen because it provides a highly controlled, empirically rich environment to practice end-to-end data reasoning. The dataset perfectly simulates a realistic business dilemma containing complex categorical variables, severe class imbalance, and hidden data leakage traps, allowing the group to demonstrate critical thinking in preprocessing and model selection without the extraneous scope creep of primary data collection.

3. Business Lens

The project will use one primary decision lens to maintain a coherent scope. Customer ranking will be treated as an operational use of the predicted response probabilities, not as a separate disconnected lens.

3.1 Primary Lens Campaign Response Prediction

The primary lens is Campaign Response Prediction. The model will estimate whether a campaign contact will result in subscription to a term deposit. The immediate model output is a probability between zero and one, followed by a class decision only when an operating threshold is required.

This lens is suitable because the response is observed historically as yes or no and the business decision occurs before allocating the next campaign call. A probability is more useful than a class label alone because it allows the marketing team to rank customers and alter the number contacted according to campaign capacity and acceptable cost.

3.2 Stakeholder Decision and Scope

Element	Proposed Definition
Primary stakeholder	Bank marketing campaign manager and campaign planning team
Decision need	Determine which eligible campaign contacts should receive priority before the call
Unit of analysis	One historical campaign contact record; not necessarily one unique customer
Prediction point	Immediately before the relevant campaign call, using only information available at that time
Primary output	Probability that the customer subscribes to the term deposit
Operational use	Rank customers or select a validation-based contact threshold
Out of scope	Automated calling, credit decisions, causal claims, and replacement of human campaign governance

A false positive lead to a call that does not produce a subscription, using staff time and potentially inconveniencing the customer. A false negative represents a potential responder who is not prioritised. The preferred trade-off cannot be fixed without information about campaign capacity, contact cost, expected value, and customer-contact policy. The project will therefore report multiple thresholds and top-ranked contact scenarios rather than presenting 0.50 as automatically optimal.

4. Problem framing

4.1 Task at hand

The proposed task is supervised binary classification. The two target classes are subscription yes and subscription no. Candidate models will generate positive-class probabilities, allowing both classification evaluation and ranking-based campaign evaluation.

4.2 Proposed Input Variables

Variable Group	Candidate Variables
Customer characteristics	age, job, marital, education
Financial relationship	default, housing, loan
Planned or known contact context	contact, month, day_of_week
Current campaign history	campaign
Previous campaign history	pdays, previous, poutcome
Economic context	emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed

The availability of each feature will be checked against the prediction point. Month, day of week, and contact channel may be used only if they represent a planned campaign schedule known before the call. Campaign-contact counts must similarly reflect information accumulated before the current decision. Economic variables will be retained initially because they were observable at the campaign time, but their strong relationship with chronological periods will require careful temporal validation.

Possible deterministic features include a previously contacted indicator derived from pdays and clearly justified groupings such as age bands for descriptive or fairness analysis. Feature engineering will remain limited and evidence based. It will not be used to create redundant variables merely to increase model complexity.

4.3 Target Variable

The target variable is y . A value of yes indicates that the client subscribed to the term deposit, while no indicates that the client did not subscribe [\[2\]](#). For modelling, these labels will be encoded as 1 and 0 respectively. The original values will be preserved in the reproducibility record.

4.4 Leakage and Feature Availability

Call duration is a major leakage risk. UCI states that duration strongly affects the output but is not known before a call is performed; after the call, the response is already known [\[2\]](#). Duration will therefore be excluded from the primary pre-contact model. A clearly labelled diagnostic experiment may include it only to demonstrate how unavailable information can inflate performance. Its results will never be used for the deployment recommendation.

Preprocessing parameters, resampling, feature selection, model tuning, and probability-threshold selection will be learned from training data only. The final test outcomes will not be used to decide which model or threshold appears best.

4.5 Class Imbalance

Positive subscriptions are expected to be the minority class. Consequently, raw accuracy may reward a model that predicts no for most records. The project will compare class weighting and, only if justified, training-fold resampling. Any resampling method will be applied inside the cross-validation pipeline so that synthetic or repeated observations cannot enter validation or test data.

4.6 Expected Project Output

- A response probability and optional threshold-based class for each evaluated record.
- A ranked customer-contact list for fixed campaign-capacity scenarios.
- A baseline and at least three alternative classification models.
- Cross-validation and untouched test results using imbalance-aware and business-oriented metrics.
- Feature-importance, error, calibration, and subgroup analyses where statistically supportable.
- An evidence-based model and contact-strategy recommendation with risks and limitations.

5. Proposed Workflow diagram

Proposed Workflow

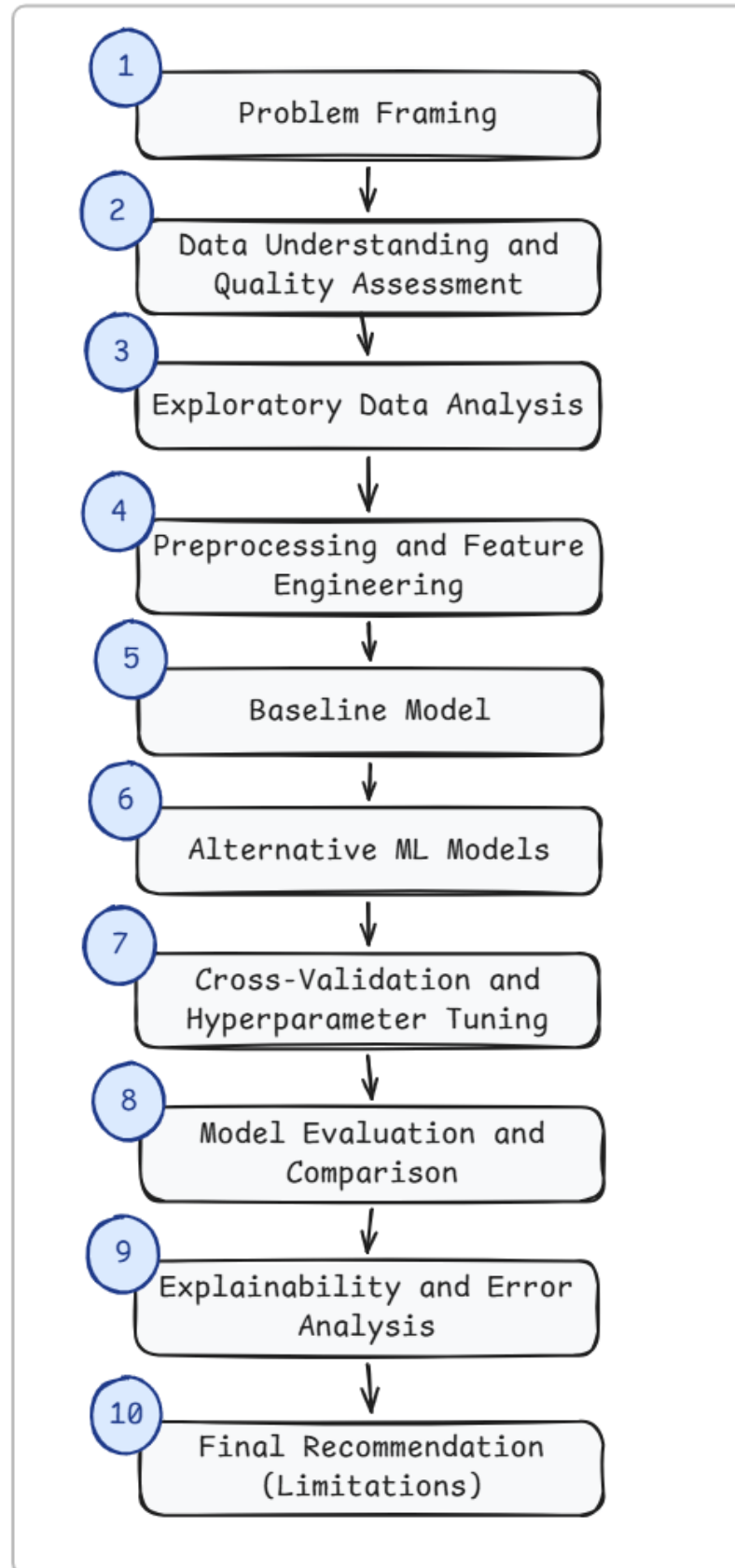


Figure 1 Proposed workflow from campaign decision to recommendation

6. Work allocation and team responsibility

The execution of this project is divided across four distinct, interdependent functional roles to ensure comprehensive coverage of the assessment rubric and clear personal accountability:

Member	Core Responsibility	Main Responsibilities
Siribaddana D. S. IT24200320	Problem Framing & EDA	Formulates the problem framing canvas, compiles the data dictionary, conducts exploratory data analysis (EDA), and documents variable distributions and insights.
Athukorala M. B IT24101937	Data Engineering	Constructs the preprocessing pipeline, ensures strict data leakage prevention (removing duration), handles categorical encoding and missing values, and maintains the preprocessing log.
Nishshanka A. D. N. N. IT24104095	Model Development	Builds the baseline model and at least three alternative machine learning architectures, executes cross-validation and hyperparameter tuning, and records structural model trade-offs.
Nawarathna M.S.G. IT24104206	Evaluation & Diagnostics	Implements evaluation metrics tailored for class imbalance (PR-AUC, F1-score), translates probability thresholds into business recommendations, and conducts error analysis diagnostics.

6.1 Shared Group Responsibilities

Although each member has a core area, major decisions will be made collaboratively. All members will review the problem framing, prediction point, decision log, preprocessing pipeline, model comparisons, interpretation of results, ethical considerations, final report, runnable notebook, three-minute demonstration, and AI-use declaration.

- Attend regular progress reviews and record agreed decisions.
- Use version control or another traceable method for shared code and documents.
- Review one another's code, written interpretation, and evidence.
- Confirm that notebook outputs match the values presented in the final report.
- Maintain an individual contribution record for the Personal Learning Journey submission.

This arrangement gives each member clear ownership while ensuring that the complete machine-learning workflow is understood by the group rather than being divided into isolated pieces.

7. Conclusion

This project proposes a supervised machine-learning approach for Campaign Response Prediction using the UCI Bank Marketing dataset. Its central purpose is to estimate, before a call, the probability that a campaign contact results in subscription to a term deposit. The probabilities will support ranking and contact-capacity analysis while maintaining one coherent primary lens.

The project will use the Guided Data Track and the richer full dataset, subject to final verification. A no-skill baseline and multiple alternative classifiers will be compared through validation designed to respect the dataset's chronological ordering. Evaluation will extend beyond accuracy to PR AUC, precision, recall, F1-score, balanced accuracy, calibration, lift, and responder capture at practical contact budgets.

Particular attention will be given to leakage, feature availability, class imbalance, repeated contacts, unknown categories, temporal change, fairness, and reproducibility. Duration will be excluded from the main model because it is not available before the call. The final recommendation will be based on evidence from the protected holdout rather than the apparent performance of a model selected in advance.

The proposed output is decision support, not an automated contact instruction or a causal estimate of marketing impact. Its value will depend on whether the selected model generalises, produces useful rankings, can be explained responsibly, and is interpreted within the limits of historical data collected in a different country and period.

8. References

- [1] Sri Lanka Institute of Information Technology. (2026). IT3091 Machine Learning Group Assignment Descriptor, Version 1. Internal module document.

- [2] S. Moro, P. Rita, and P. Cortez. "Bank Marketing," UCI Machine Learning Repository, 2014. [Online]. Available: <https://doi.org/10.24432/C5K306>.